Literal equations and formulas

1

a $x = y - z$

b $x = 4y$

c $x = -\dfrac{y}{8} - 7$

d $x = 45 - \dfrac{9n}{2}$

e $x = -45 - \dfrac{9n}{5}$

f $x = \dfrac{m}{k - n}$

2

a $m = ghy$

b $m = \dfrac{y + 5}{9x}$

c $m = -\dfrac{x}{2k} - n$

d $m = \dfrac{x}{3k} - n$

3

a $k = \dfrac{3r}{r - 1}$

b $k = \dfrac{7\left(m + 3\right)}{5x}$

4 $y = \dfrac{x}{h - k}$

5

a $R = \dfrac{V + E}{I}$

b $R = \dfrac{D}{T}$

6 Possible answers:

- Subtract 32 from both sides, then multiply by 5 and divide by 9. This results in $C = \dfrac{5\left(F - 32\right)}{9}$.
- Multiply all the terms by 5, then subtract 160 from both sides. The last step is to divide by 9. This results in $C = \dfrac{5F - 160}{9}$.

7 $T = \dfrac{360B}{\pi A} - 2R$

8 $v_0 = \dfrac{S - \frac{1}{2}at^2}{t}$ or $v_0 = \dfrac{S}{t} - \dfrac{1}{2}at$

9 $v_f = at + v_i$

10 $30^\circ C$

11

a 57.6 gigabits

b 28.8 gigabits/hour

12 $h = 12$

13 $h = 9$

14 $h = 10$

15 $I = 0.25$

16 $a = -10$

17 $b = 21.60$

18 $R = 2C$

19 160 cm^3

20 $280\pi \text{ cm}^2$

21

a $x = \dfrac{y - 73}{0.28}$

b $x = 78.571$

c 2049

22

a $x = \dfrac{y - 1.37}{0.106}$

b $x = 20.094$

c 2021

Additional questions

23 We can have multiple equations solving for various unknowns. Once these equations have been rewritten, we do not have to do this manipulation again.

25 $l = \dfrac{P - 2w}{2}$

26 $C = \dfrac{5(F - 32)}{9}$

27

a $r = \dfrac{d}{t}$

b 20 mi/h

28 Both equations can be rearranged for x. When $3x + 8 = 2$ is rearraged for x the answer is numerical. Whereas when $3x + y = z$ is rearranged for x the answer has unknowns, y and z, in it. However if you substitute $y = 8$ and $z = 2$, you get the same numerical answer as solving $3x + 8 = 2$.

29 Rufino has an error in his working.

$$\begin{aligned} KE &= \frac{1}{2}mv^2 && \text{The equation for } KE \text{ as given} \\ 2KE &= mv^2 && \text{Multiplying both sides by 2} \\ \frac{2KE}{v^2} &= m && \text{Now he should have divided by } v^2 \\ m &= \frac{2KE}{v^2} \end{aligned}$$